

Assignment No - 01

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Course : CSE 320

Section : 20.

Ans to the Q No - 1

The communication between the two friends most likely falls into the Application Layer of the OSI model. To communicate a process sends a request to the other process and receives a response. Process to process communication is the duty of the application layer. Here the conversation revolves around exchanging ~~par~~ 'parcels' representing data or goods. This aligns with the Application Layer's function. The friends discuss the number of parcels and receiving readiness, not technical aspects like network addresses, protocols or physical delivery method. It indicates that it is within the application environment itself. A negotiating quantity doesn't directly correspond to Transport layer functions like segmentation. So it falls ~~within~~ within application layer.

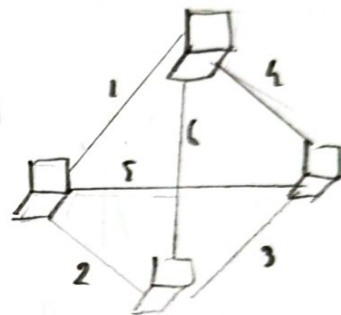
Ans to the Q No-2

For n devices in network,

(i) Mesh Topology: In a mesh topology, every device is connected to every other the devices. For n devices, number of links is $= \frac{n(n-1)}{2}$
for ~~full~~ duplex ~~and~~ mode but for simplex it's $n(n-1)$

For example, there are 4 devices,

$$\begin{aligned}\text{So, number of links} &= \frac{4(4-1)}{2} \\ &= 6\end{aligned}$$

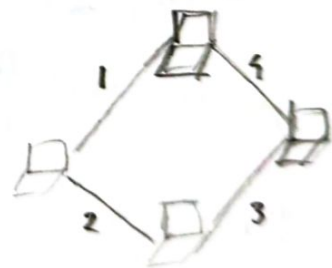


(ii) Ring topology: In a ~~ring~~ ring topology each device is connected with two other devices,

Number of links will be $= n$

For example, there are 4 devices,

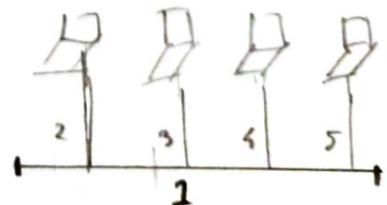
So, ~~new~~ number of links $= 4$



(iii) Bus topology: In a bus topology all devices share a single line.

Number of links $= n+1$

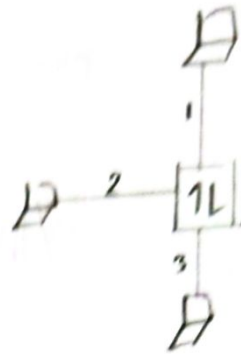
For example, if there are 4 devices,
number of links $= 5$



(2)
iv) Star topology: In a star topology, all devices are connected to a central hub or switch.

So, number of links = $n - 1$

If there are 4 devices, then,
number of links = $4 - 1 = 3$.



Ans to the Q No-3

There are five elements of data communication. They are-

i) Sender:- Sender is the device that sends the data message. It can be from computer, telephone, mobile, video ^{camera} ~~camera~~ etc.

ii) Receiver:- The receiver is the device that receives the message. It can be computer, mobile, telephone and so on.

iii) Message- The message is the data or information which is communicated. Information or data includes text, numbers, pictures, audio and video.

iv) Medium: Medium is the path by which message travels from sender to receiver. Some examples of medium are twisted pair wire, fibre optic cable, etc.

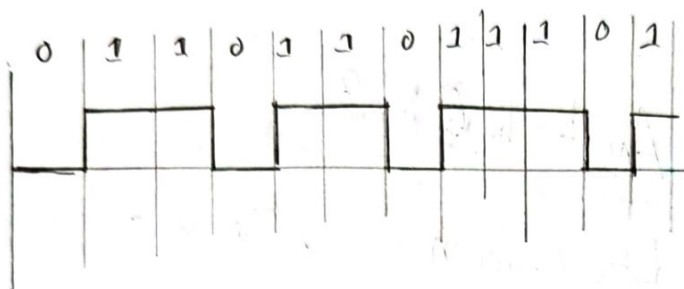
v) Rules & Protocol: Protocol is set of rules that governs data communication.

It ensures that the sender and receiver can understand each other and interpret the message correctly.

Ans to the Q No-4

(a)

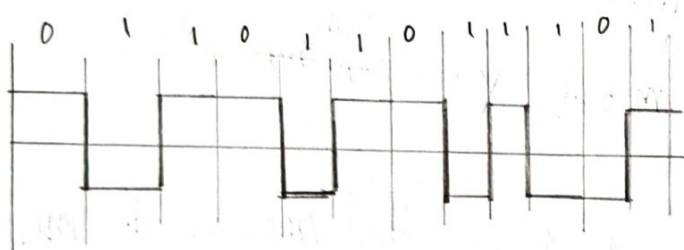
NRZ



0 → 0

1 → +ve

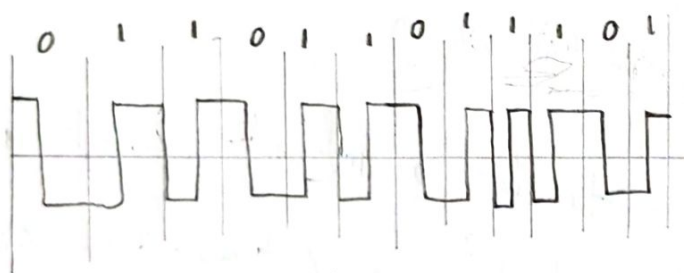
NRZ-I



0 → no invert

1 → invert

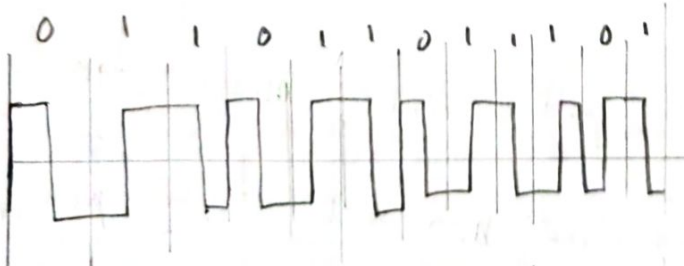
MAN



0 → 1

1 → 0

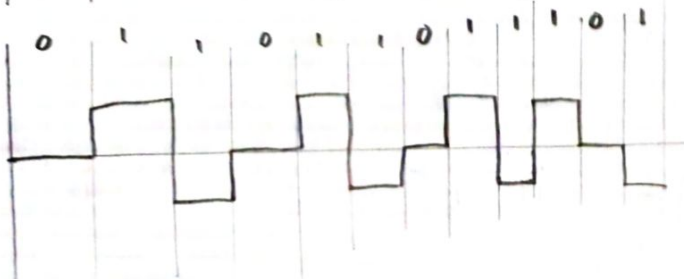
DMAN



0 → invert

1 → no invert

AMI



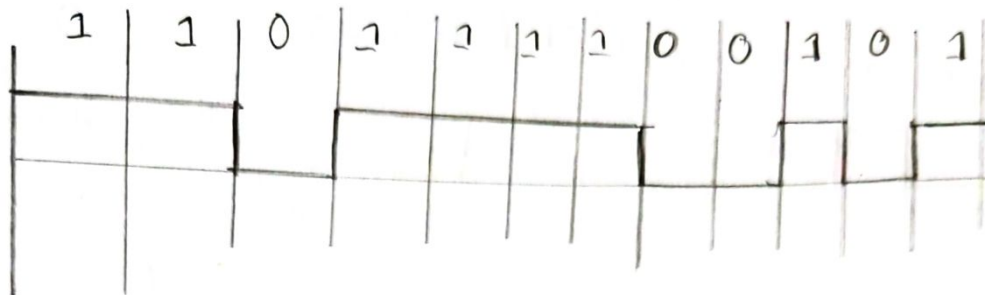
0 → 0

1 → +ve, -ve

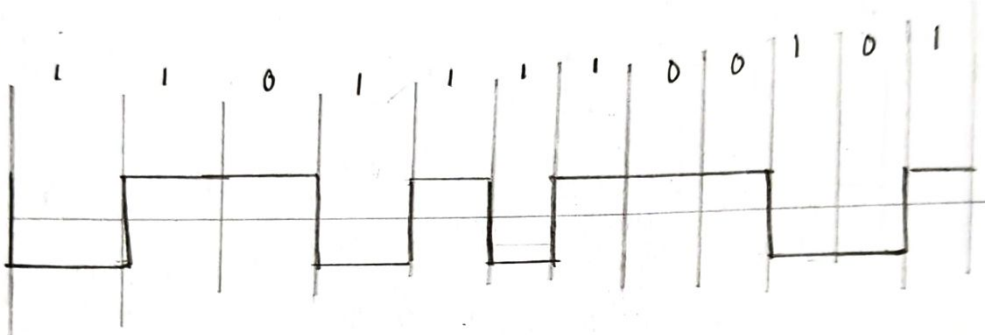
3)

b)

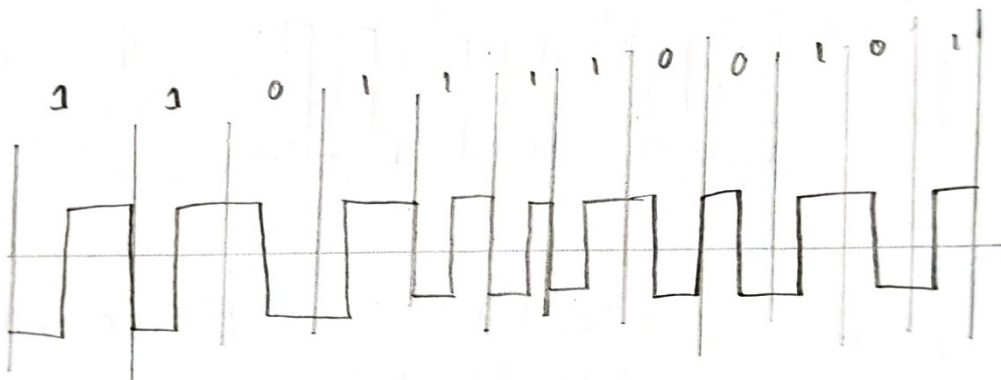
NRZ



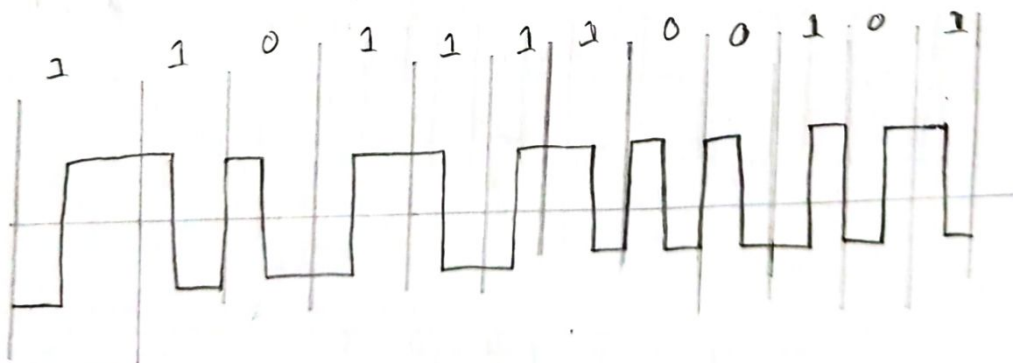
NRZ-I



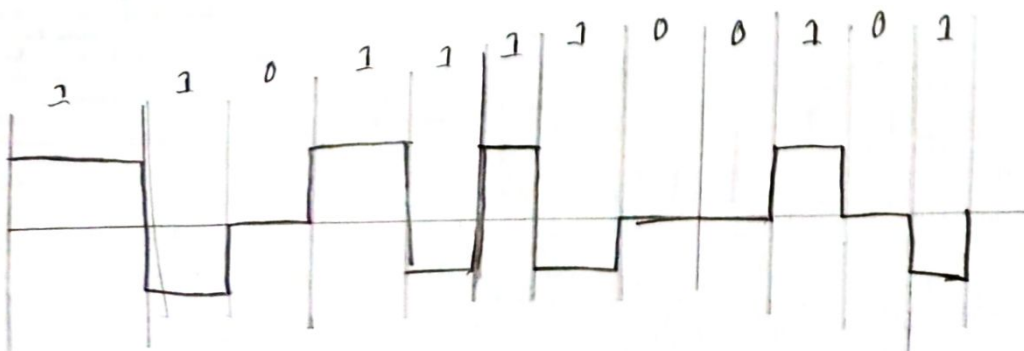
MAN



DMAN

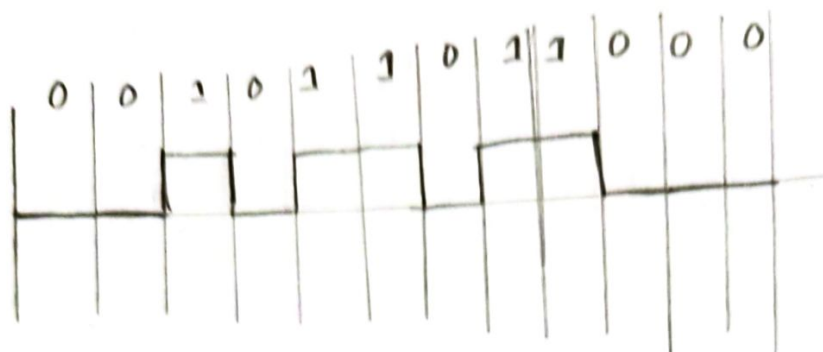


AMI



(C)

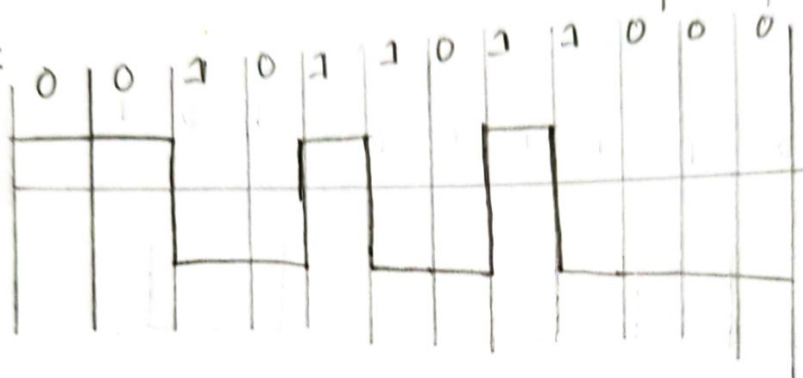
NRZ



0 $\rightarrow$ 0

1 $\rightarrow$ +ve

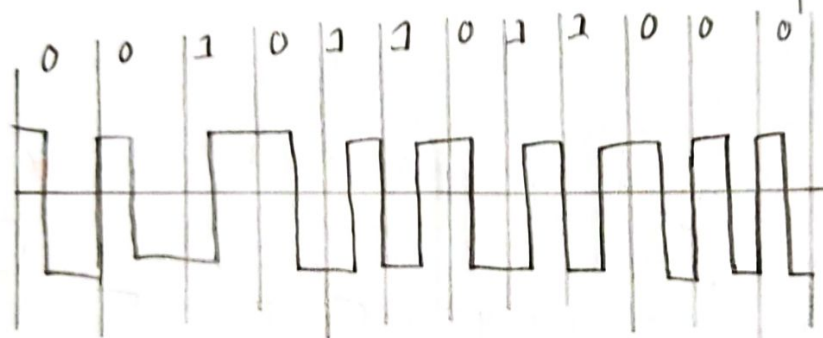
NRZ-I



0 $\rightarrow$ no invert

1 $\rightarrow$ invert

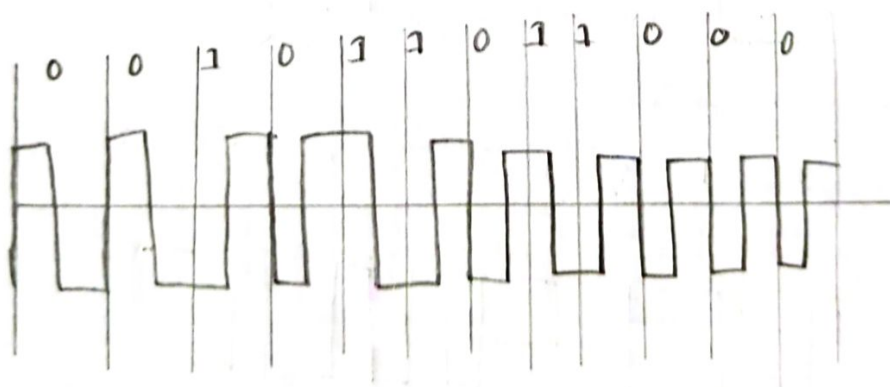
MAN



0 $\rightarrow$ L

1 $\rightarrow$ H

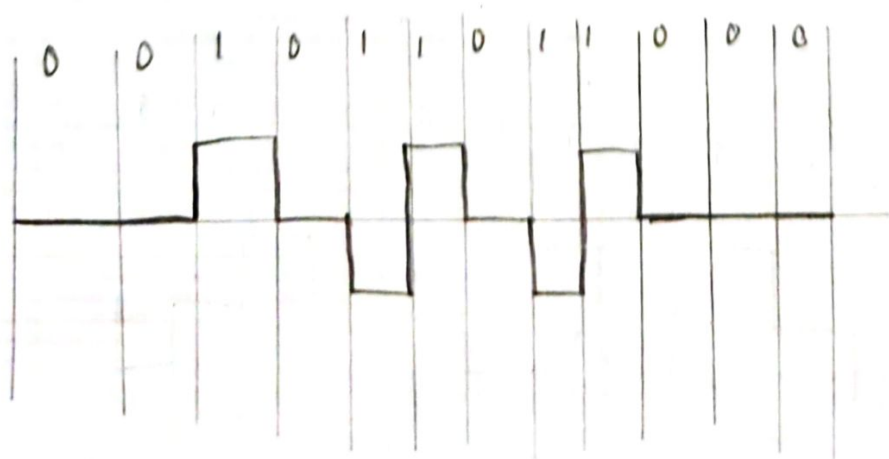
DMAN



0 $\rightarrow$ invert

1 $\rightarrow$ no invert

AMI

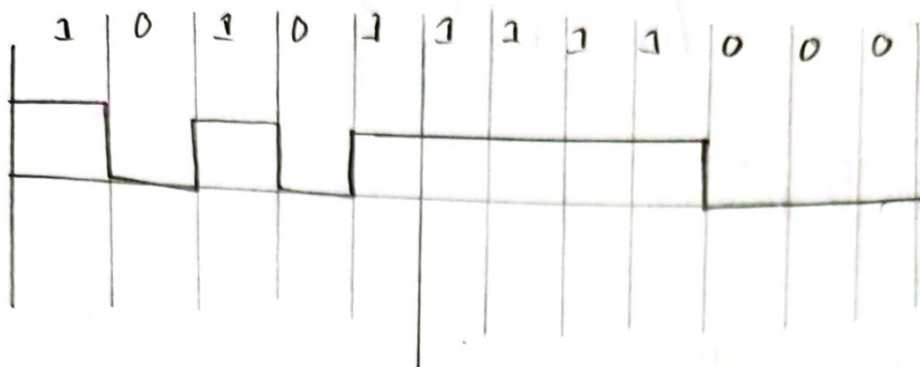


0 $\rightarrow$ 0

1 $\rightarrow$ +ve, -ve

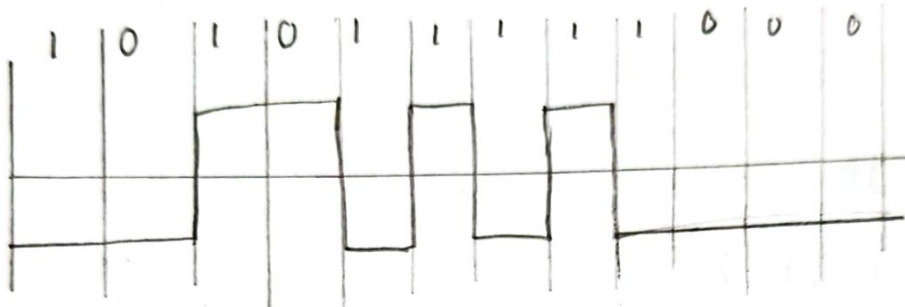
(d)

NRZ



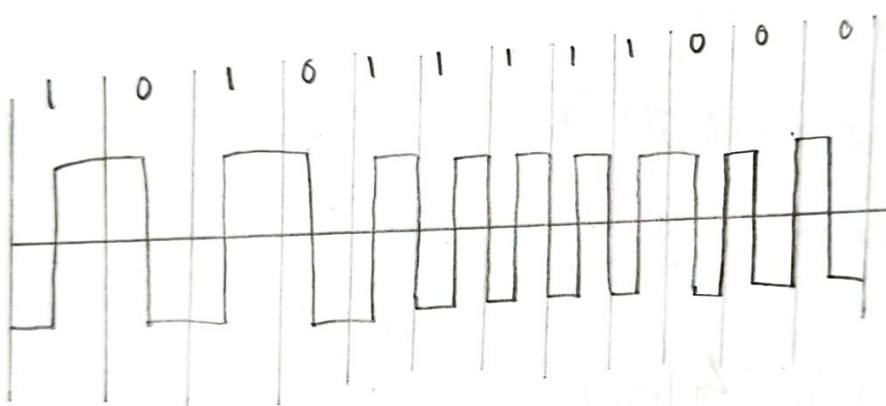
0 → 0
1 → +ve

NRZ-I



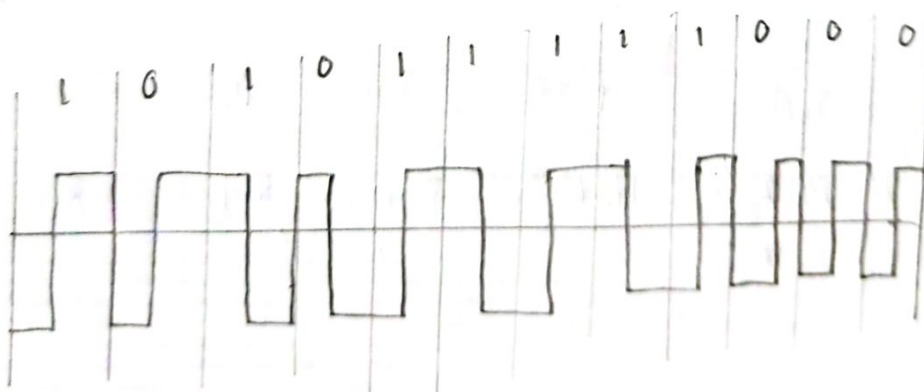
0 → no inv
1 → inv

MAN



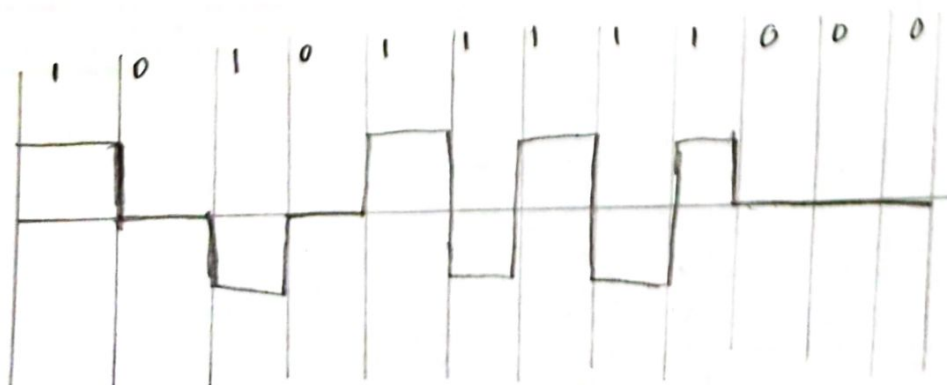
0 → L
1 → H

DMAN



0 → inv
1 → no inv

AMI



0 → 0
1 → +ve, -ve

Ans to the Q No-5

(a) There are 5 networks.

(b) There will be 2 hops.

D. port : 25

S. port : 60010

D. IP : 111.121.12.2

S. IP : 111.121.12.0

D. MAC : 2C:9A:1C:94:5C:E3

S. MAC : 2C:54:91:88:C9:E3

(c) There will be 4 hops

D. Mac	S. Mac	D. IP	S. IP	D. Port	S. Port		
2C:54:91:00 :9A:1C:D	9A:54:210:68 :C9:2C	172.67.13.225	53.103.4.80	50100	80	Data	Trailer

Drawbacks of line coding are given below - Ans to the Q No-6

i) Baseline Wandering - Baseline wandering is the running average of received signal power. A long strings of 0 and 1 can cause a drift in the baseline making it difficult for the receiver to decode properly.

ii) DC Components - DC components occurs when the voltage level in a digital signal is constant for a while, the spectrum creates very low frequencies. It can lead to baseline wandering and distort the signal, making it difficult for receivers to interpret.

iii) Self-synchronization - Self synchronization is the ability of a communication system to recover the clock signal at the receiver without relying on an explicit clock signal transmission. It enhances the accurate interpretation of the received signal.

⑩ Built in Error Detection: Built in error detection involves incorporating mechanisms within the communication system to identify errors during signal transmission.

⑪ Noise & Interference: Noise and interference can distort signals, leading to misinterpretation and errors in data recovery.

⑫ Complexity: Complexity refers to the intricacy and sophistication of the communication system and its components. Highly complex systems may introduce challenges in implementation, maintenance, and achieving successful signal interpretation.

Ans to the Q No-7

(A) ~~Network layer~~ Network layer

(B) Physical layer

(C) Application layer

(D) Transport layer

(E) Data link layer

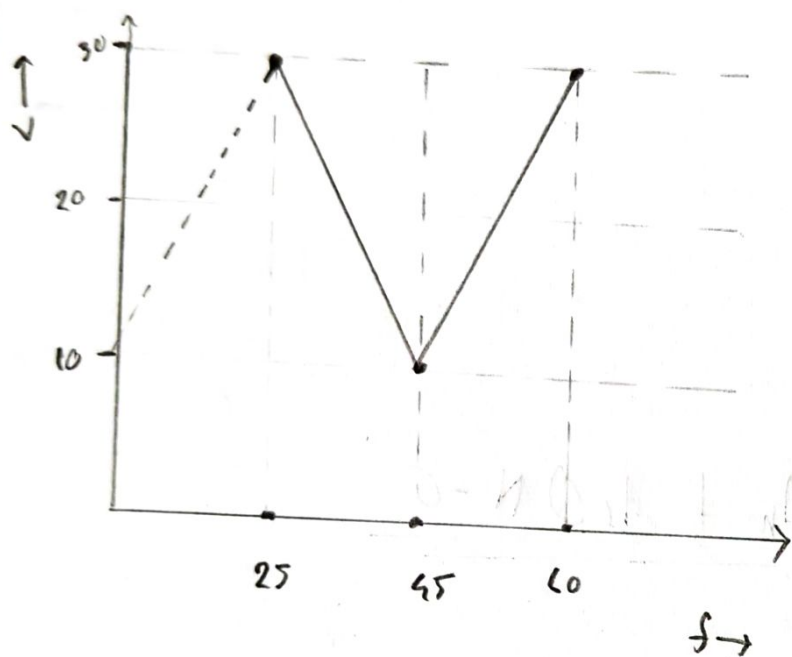
(F) Application layer

Ans to the Q No-8

(a) A non periodic composite signal is the signal that does not repeat in a perfect pattern in time. Composite signals are composed of multiple frequencies combining various sinusoidal and non-sinusoidal components.

(b) Bandwidth = highest frequency - lowest frequency
= $(60 - 25) \text{ kHz}$
= 35 kHz .

(c) The frequency spectrum is given below,



Ans to the Q No-9

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$S_o, X = 44 \text{ MHz}, Y = 22 \text{ mW}, Z = 18$

(a)

$$\text{Bandwidth} = 44 \times 10^6 \text{ Hz}$$

$$\text{SNR} = 18$$

$$\begin{aligned} \text{data rate} &= \text{Bandwidth} \times \log_2(1 + \text{SNR}) \\ &= 44 \times 10^6 \times \log_2(1 + 18) \\ &= 186.91 \text{ Mbps.} \end{aligned}$$

(b) $\text{data rate} = 186.91 \times \frac{2}{3} = 124.61 \text{ Mbps.}$

Now,

$$124.61 = 2 \times 44 \times \log_2 L$$

$$\Rightarrow \log_2 L = 1.42$$

$$\Rightarrow L = 2^{1.42}$$

$$\Rightarrow L = 2.67.$$

As asked for better error performance, so the signal level should be 2.

(C)

$$\text{bandwidth}^L = 2 \times 44 \text{ MHz}$$

$$= 88 \text{ MHz.}$$

So,

$$186.91 = 2 \times 88 \times \log_2 L$$

$$\Rightarrow \log_2 L = 1.061$$

$$\Rightarrow L = 2.086.$$

$$\Rightarrow L \approx 3$$

So in order to maintain same data rate signal level should be made to 3.

Ans)

Ans to the Q No-10

(a)

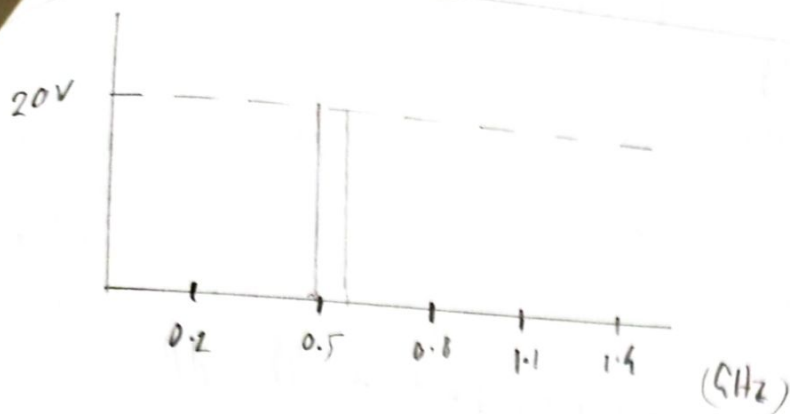
$$\text{cycle} = 8$$

$$T = 16 \text{ ns} = 16 \times 10^{-9} \text{ s.}$$

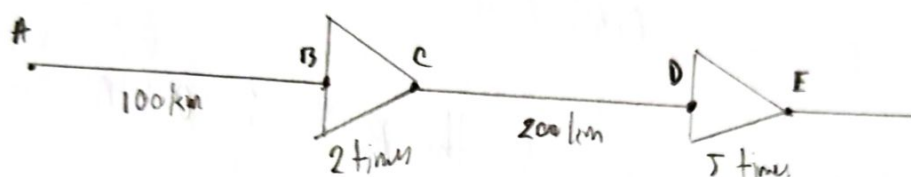
$$\text{time period} = \frac{16 \times 10^{-9}}{8} = 2 \times 10^{-9} \text{ s.}$$

$$\text{frequency} = \frac{1}{2 \times 10^{-9}} = 0.5 \times 10^9 \text{ Hz}$$

$$= 0.5 \text{ GHz.}$$



(b)



From A to B,

$$\text{power loss} = 100 \times 5 = 500 \text{ kW/km}$$

$$\begin{aligned} \text{signal power} &= (5 \times 10^6 - 500 \times 10^3) \text{ W} \\ &= 4.5 \times 10^6 \text{ W} \end{aligned}$$

$$\begin{aligned} \text{Attenuation} &= 10 \log_{10} \frac{4.5 \times 10^6}{5 \times 10^6} \\ &= -0.357 \text{ dB} \end{aligned}$$

$$\begin{aligned} \text{From B to C} &= \left(10 \log_{10} \frac{2P_2}{P_1} \right) \\ &= 3.01 \text{ dB} \end{aligned}$$

$$\begin{aligned} \text{From C to D} &= (200 \times 0.05) \\ &= -10 \text{ dB} \quad [\text{loss rate}] \end{aligned}$$

$$\text{from D to E} = 10 \log_{10} \left(\frac{P_2}{P_1} \right)$$

$$= 6.99$$

Now,

$$\text{total change in power} = (-0.457 + 3.01 - 1.0 + 6.99) \text{ dB}$$

$$= -0.457 \text{ dB}$$

As the answer is negative, so the power is attenuated.

Ans)